7.1 Web forms

In the previous chapter, you took the first steps toward having self-serve user features in RiffMates. And although it wasn't explicitly called out, you started using *web forms*. The login page is a web form with two fields: Username and Password. Submitting this form gets handled by a built-in Django view, which adds data to the user's session, indicating they are authenticated. The mechanisms the login view uses are built on top of the tools available to you as a programmer. With them, you can build your own forms and have users submit data to your site.

The connection between a web form's HTML and Django is loosely coupled. Django provides utilities to reduce the amount of coding you need, but they are not required. The Django `Form` object provides a shortcut for creating web forms and their corresponding HTML, but the processing of data submitted through this form is generic, and it is up to you to code your view as you see fit.

An instance of a Django `Form` specifies a number of fields, with each field corresponding to an HTML tag. You can use a form instance in a template to render the HTML tags, but you don't have to. Figure 7.1 shows a form with a field for entering a person's name and email address along with the corresponding HTML.

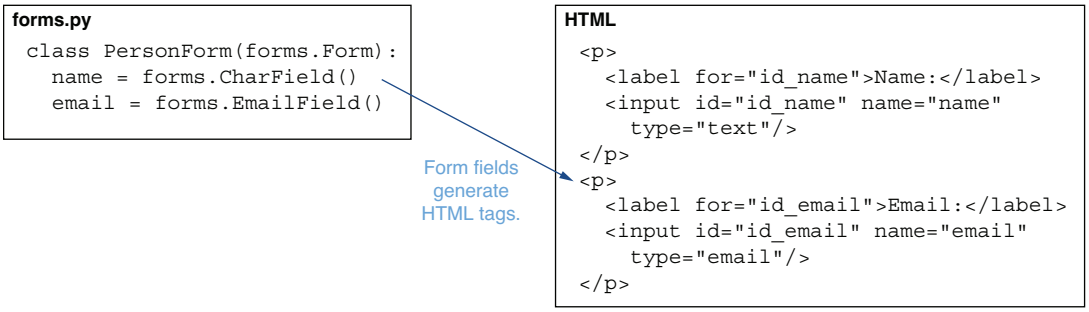


Figure 7.1 The relationship between a `Form` object and the HTML

Form fields get rendered through an associated widget object. The widget object is responsible for rendering the corresponding HTML tag. There are over 20 different widgets provided by Django, but most of them result in the `<input>` tag, with varying values of the `type` attribute. For example, the `CharField` renders as `<input type="text">`, while the `EmailField` renders as `<input type="email">`. Beyond the `<input>` tags, there are also widgets for drop-down selects, multiple selection boxes, radio buttons, dates, times, and file handling.

You have some control over the appearance of the widget, but you can't use template features, like conditional formatting. This limitation can be a little frustrating; a